

PROGRESS REPORT • SESSION 2

A Similarity Index for Cellular Automata

Measuring behavioral similarity between CA rules, from exact classification to a continuous, quantitative index

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Why "similarity" between rules?

Conway's Game of Life has a large family of “life-like” rules — different birth/survival sets that produce visually similar behavior.

From random initial configurations these rules often look indistinguishable: it is only specific small patterns (gliders, oscillators, still lifes) that reveal a difference.

This raises the core question of the project:

How similar are two CA rules, based on their behavior — and can that similarity be measured, not just judged by eye?

SCOPE

- Primarily 1-dimensional CA rules (elementary CA and generalizations)
- Extended this session to a 2-dimensional case: Conway's Game of Life and life-like variants
- Two complementary questions: are two rules exactly the same? and if not, how different are they?

Three methods, one question

01

Exact / inheritance relations

Binary test: do two rules show exactly the same behavior? Built from state permutation, state/neighbor addition-removal.

02

Evolution graph

Continuous measure: canonical form of the transition graph via adjacency-matrix eigenvalues, compared by cosine distance.

03

Spectral moments

Fixed-length cycle-length signature per rule — solves the eigenvalue method's variable-vector-length limitation.

Each method was also stress-tested on the 2D case: Conway's Game of Life and life-like variants.

Four filters for exact equivalence

Rule 0 and the highest rule are always trivially the same, regardless of neighbors, states or dimensionality. Beyond that trivial case, we found 4 transformations that preserve a rule's exact behavior:

State permutation

A bijection between two rules' state sets that preserves every transition. Example: swapping $\{0,1\} \rightarrow \{a,b\}$ identifies rule 01,10 and 31,13 as the same rule wearing different labels.

Mirror / inverse rules

Reversing the order of the neighbor positions (a left-right spatial reflection) maps a rule onto an equivalent one — a spatial symmetry of the same behavior.

Neighbor deletion

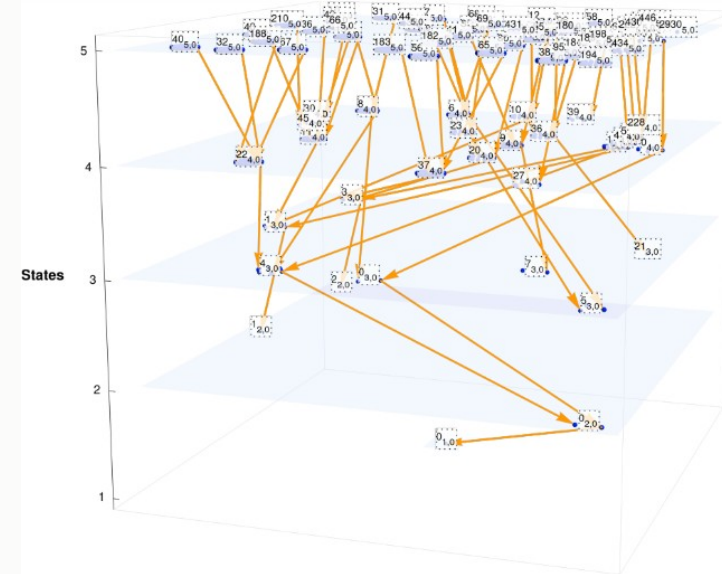
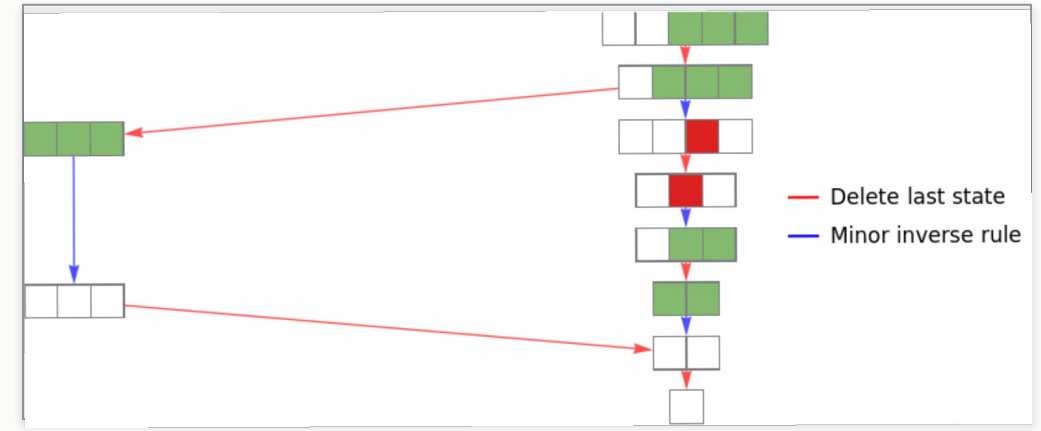
If the output never actually depends on one neighbor position, that position can be dropped — reducing the neighborhood size without changing behavior.

State deletion

If the top state's only transition is to state 0, it can be removed, moving the rule to a smaller state space. The reverse (state addition) also holds.

Moving across rule spaces

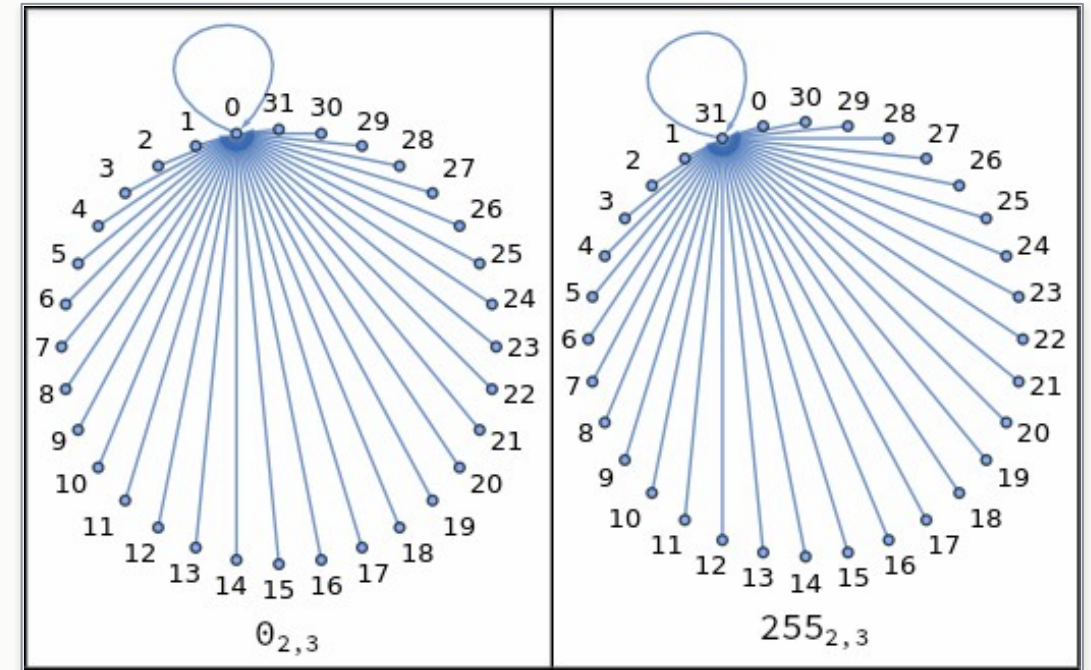
- A second relation: adding a state that transitions to 0 moves a rule from an n -state space to an $(n+1)$ -state space without changing its behavior.
- The reverse (removing a state) only works when 2 structural conditions hold — when it fails, the target rule can still often be reached via the state-permutation graph instead.
- Chaining both relation types over all tested small spaces produces a connection graph: most paths converge toward the trivial rule, as expected for dimensionless CA.



Delete-last-state and minor-inverse-rule relations chained together

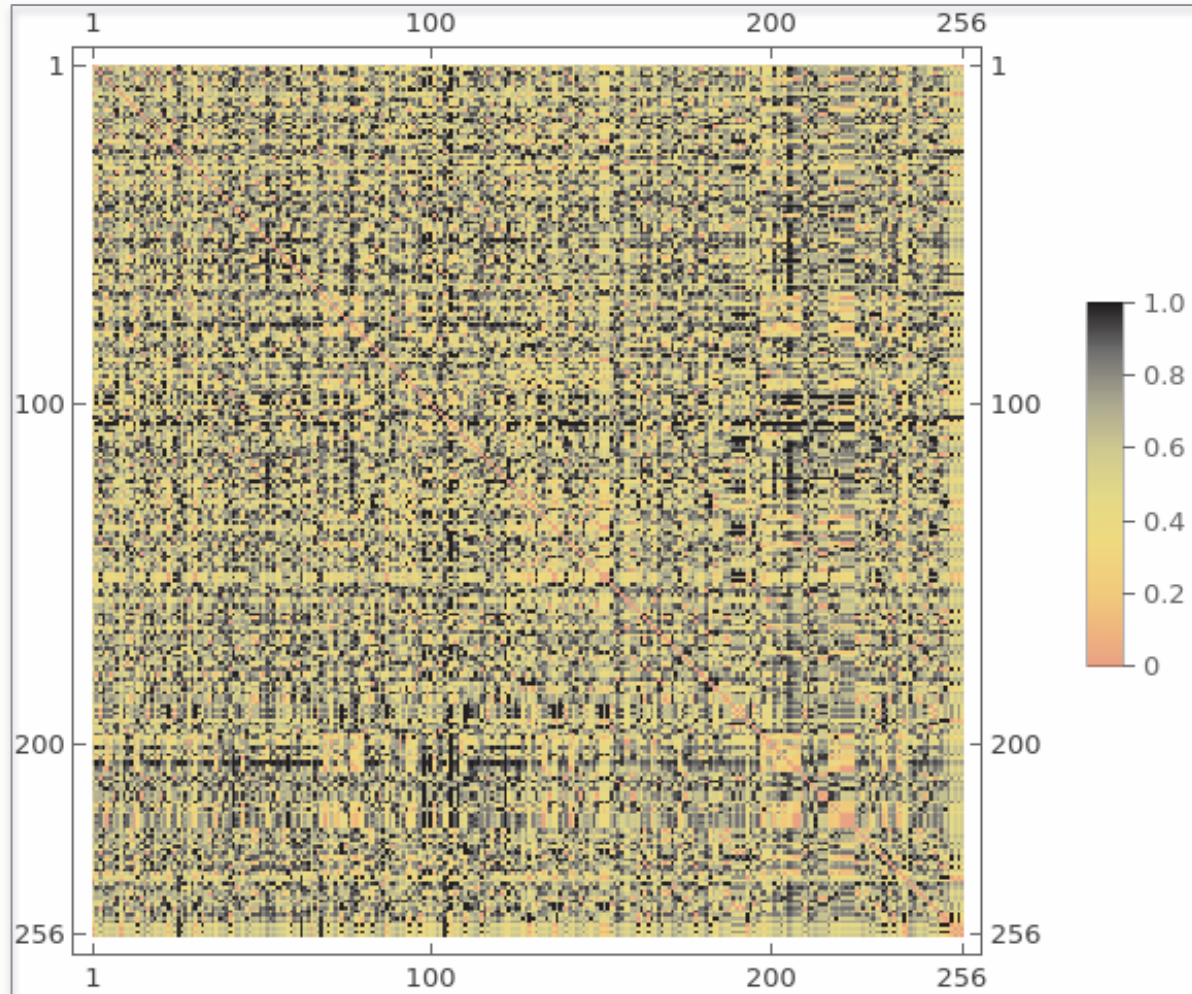
From binary match to continuous distance

- Evolve a rule over every possible finite configuration of a space: each configuration is a node, each transition an edge — this is the rule's evolution (transition) graph.
- Two graphs that are isomorphic (same shape, different labels) mean the rules are behaviorally equivalent, generalizing method 1.
- To go beyond yes/no: take the eigenvalues of each graph's adjacency matrix as a canonical, label-free feature vector, and compare rules with cosine distance between eigenvalue vectors.
- Distance 0 $\Rightarrow$ equivalent rules; small distance $\Rightarrow$ similar dynamics (e.g. rule 8 behaves like rule 0, just slower).



Rule 0 and rule 255 (5-cell space): isomorphic transition graphs

Similarity across all 256 elementary CA



TWO OPEN PROBLEMS

- Vector length problem: the eigenvalue vector's length equals the graph's node count, so rules can only be compared directly within the same space size.
- Space-size problem: larger spaces reveal more unique behaviors, but with diminishing returns — what size is “enough” is not obvious.

→ Addressed next by rotation-class graphs and spectral moments.

A cleaner canonical form — and a debunked artifact

- Alternative construction: quotient the graph by cyclic rotations of each configuration (e.g. 0001, 0010, 0100, 1000 collapse to one node). Well-defined because the rule is homogeneous and the boundary is cyclic.
- Node count now depends only on space size, not on the rule — a step toward comparable vector lengths.

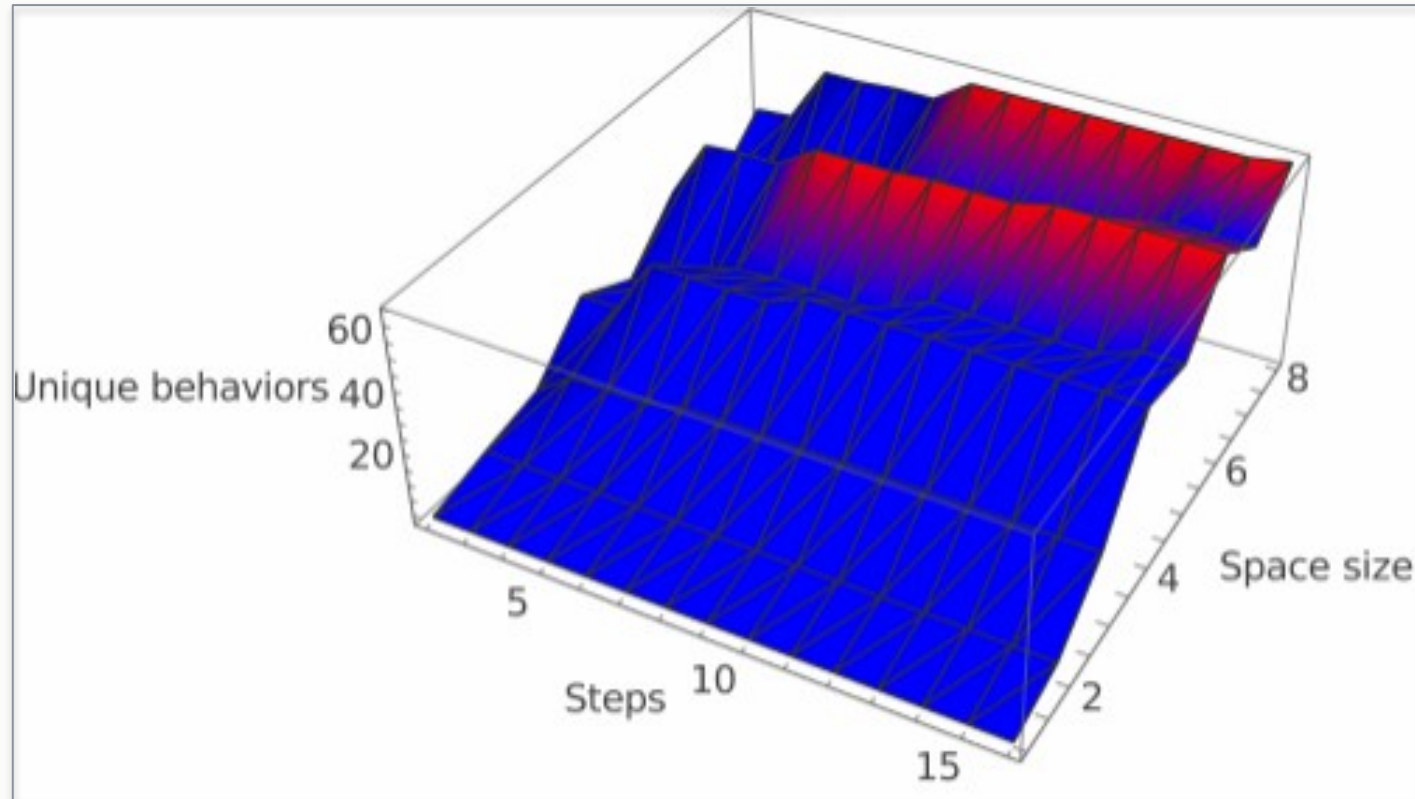
FINDING: rules 15, 85, 170, 240 (+ 30 as control)

- Raw graph (MakeGraph): all 4 rules merge together — but only at even space sizes (2,4,6,8); at odd sizes only 15~85 and 170~240 merge.
- Rotation-class graph (MakeGraphRot): the even/odd split disappears — at every tested size, only 15~85 and 170~240 merge, never all 4.
- Conclusion: the full 4-rule merge was an artifact of parity in the raw construction, not a real property of the rules.

A fixed-length behavioral signature

- Definition: evolve a rule j steps from every possible configuration, and let m_j be the fraction of configurations that returned to their starting configuration after exactly j steps.
- The result is a vector $(m_1, m_2, \dots, m_j)$ of fixed length $k_{\max}$ — independent of the rule and of how many nodes its graph has.
- This directly resolves the eigenvalue method's vector-length problem, letting rules from different spaces (state counts, neighborhoods, even dimensions) be compared on equal footing.
- Rule similarity is again cosine distance, now between spectral-moment vectors instead of eigenvalue vectors.

How many steps and how much space are enough?



Unique behaviors found, as a function of step count and space size (ECA)

- Repeating the similarity-matrix experiment while varying step count (2–16) shows how many distinct behaviors the method can tell apart.
- Sweeping space size at the same time locates the region (in red) where more than 60 unique behaviors are resolved — i.e. where the method has “enough resolution” to be useful.

Does the method generalize to 2 dimensions?

- Same two constructions (functional-graph quotient, spectral moments) applied to Conway's Life on a toroidal grid with a Moore (8-cell) neighborhood.
- Symmetry quotient generalizes naturally: instead of cyclic rotations (1D), we quotient by toroidal translations in x and y.
- Implementation validated against known Life patterns first: a block (still life) stays fixed, and a horizontal blinker correctly rotates to vertical.
- Test bed: 4x4 toroidal grid ($2^{16} = 65,536$ configurations — the same scale already validated at 1D size 16), comparing standard Life (B3/S23) against two variants: Life+S8 (B3/S238) and Life+B8 (B38/S23).

Life, Life+S8, Life+B8: cosine distance

Rule A	Rule B	Distance
Life (B3/S23)	Life+S8 (B3/S238)	0.000516
Life (B3/S23)	Life+B8 (B38/S23)	0.000000
Life+S8 (B3/S238)	Life+B8 (B38/S23)	0.000516

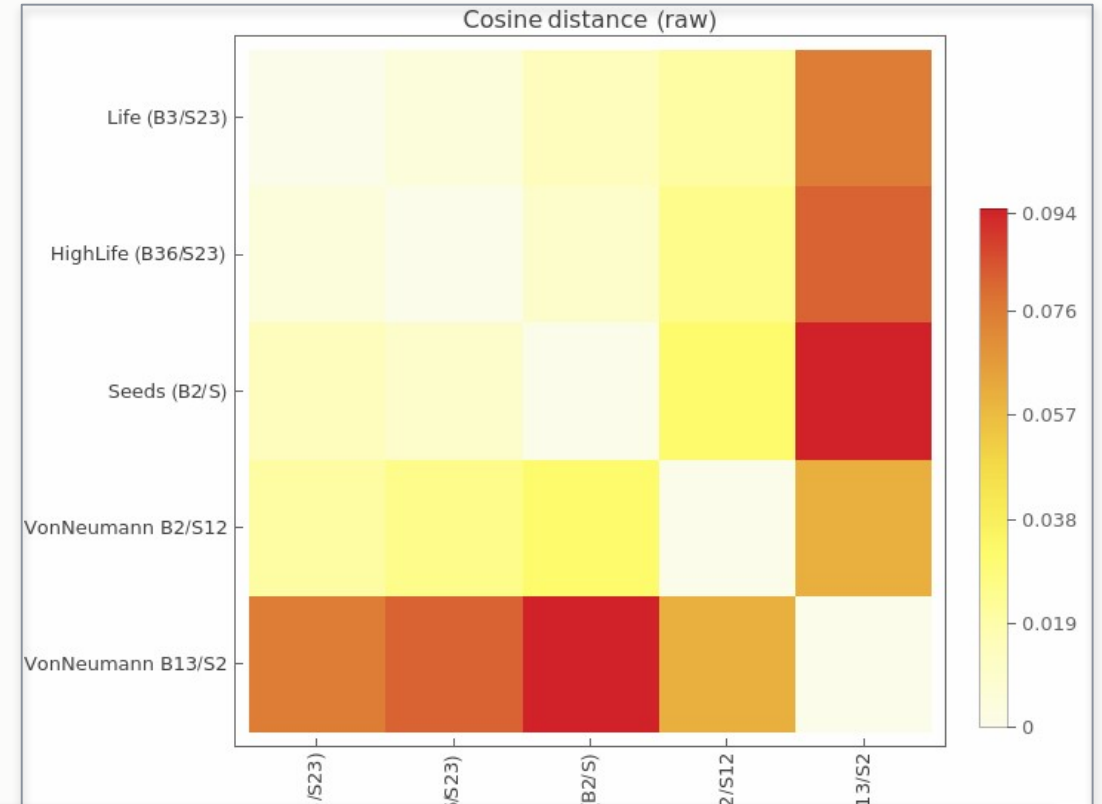
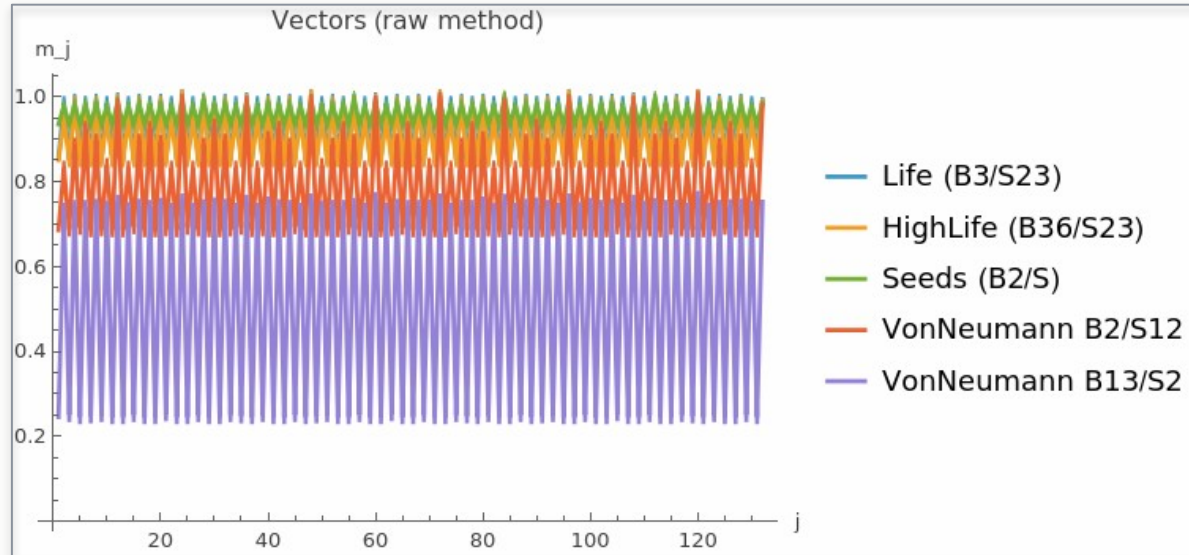
KEY FINDING

Life and Life+B8 are indistinguishable at this grid size (distance 0): reviving a cell only when all 8 neighbors are alive never changes the graph's cycle structure once quotiented. The survival-side change (Life+S8) has a small but non-zero effect — suggesting survival rules matter more structurally than birth rules here, on a 4x4 toroidal grid.

Scaling up: when exact enumeration breaks

- At 6x6 (36 cells), exact enumeration is infeasible: $2^{36} \approx 68.7$ billion raw states, and even the translation-quotiented graph has ≈ 1.9 billion nodes — neither fits in memory.
- Solution: Monte Carlo estimation. Sample random configurations, run a transient so each settles onto its attractor, then measure recurrence time — the fraction of samples recurring within j steps estimates m_j directly.
- Sampling parameters: 3,000 samples per rule, 200-step transient, 150-step cap to detect a period (unresolved beyond that).
- Tested 5 rules: 3 Moore-neighborhood, life-like rules (same family as Life) and 2 Von Neumann-neighborhood rules (4 neighbors — outside the life-like family by definition), across both graph constructions (raw and translation-quotiented).

Neighborhood type, not the B/S set, drives similarity



The 3 Moore-neighborhood (life-like) rules cluster tightly (cosine distance ≈ 0.0001 – 0.0065), while both Von Neumann rules sit far away (≈ 0.05 – 0.09). Both graph constructions agree closely except for VonNeumann B13/S2 — also the only rule with $\sim 24\%$ of samples unresolved within the step cap, hinting at long-period or chaotic dynamics.

Where the project stands

- Built a full pipeline from exact behavioral equivalence (state permutation, state/neighbor relations) to a continuous similarity measure (eigenvalue vectors, then spectral moments).
- Spectral moments solve the eigenvalue method's key limitation — fixed vector length — enabling comparison across spaces of different size, state count, and even dimensionality.
- Rotation/translation-class graphs give a cleaner canonical form and correctly identified a parity artifact in an earlier finding (the {15,85,170,240} “full merge”), replacing it with the real result: only 15~85 and 170~240 merge.
- Extended and validated the whole approach in 2D, up to a 6x6 grid via Monte Carlo sampling

Toward the research paper

- Consolidate the three methods and their findings into a single manuscript, strengthening the theoretical grounding behind each (formal definitions, proofs of well-definedness for the quotient constructions, error bounds for Monte Carlo sampling).
- Push the spectral-moments comparison to larger, systematic sweeps (more rules, more space sizes) now that step/space resolution has been characterized.
- Extend the 2D Monte Carlo study beyond 5 rules to a broader life-like/non-life-like sample, to test how robust the neighborhood-dominates-similarity finding is.